

Khaled Lakhdher

Computer Engineer | AI & Data Science | Full-Stack AI Developer

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PROFILE

Passionate Machine Learning engineering student with a solid foundation in deep learning, computer vision, and natural language processing. Proficient in Python and TensorFlow, with hands-on experience building and deploying AI models. Eager to apply my skills to real-world challenges and contribute to innovative artificial intelligence solutions.

EXPERIENCE

AI Engineer — Final-Year Project (PFE)

Feb 2026 – June 2026 · Tunisia

CTT – Carthage Training & Technologies

- Designed and developed **Hotel Mind**, a conversational hotel-booking assistant on a grounded RAG architecture connected to a real-time SOAP API, with a hallucination-mitigation layer that verifies every factual claim.
- Delivered a containerized full-stack solution reaching nDCG@5 0.82, MRR@5 0.90, and a 100% success rate on a benchmark query set.

Artificial Intelligence Developer Intern

July 2025 – September 2025 · Hammamet

CTT – Carthage Training & Technologies

- Fine-tuned a Hugging Face NLP model to classify customer reviews by sentiment (positive, negative, neutral).
- Built a Django backend to manage customer data and automate export of results to Excel for reporting.
- Handled all preprocessing, training, and integration scripts in Python.

PROJECTS

Nexus AI — AI-Employee Management Platform

- Built a multi-tenant platform (FastAPI, PostgreSQL/pgvector, Redis, Next.js) governing LLM agents with IAM permissions, vector memory, multi-agent orchestration over a Redis message bus, and real-time WebSocket monitoring, behind a provider-agnostic model gateway (Anthropic, OpenRouter, Gemini).

Personalized Voice Assistant

- Fine-tuned Whisper (STT) and SpeechT5 (TTS) with CSM-1B voice cloning to build a personalized voice assistant, cutting response time by 30%.

AI Image Colorization & Automatic Code Generation

- Trained a Pix2Pix GAN (U-Net / PatchGAN) for automatic image colorization and built a multi-agent system (Planner / Architect / Coder) that generates full-stack projects (Flask, HTML/CSS/JS, APIs) from a single prompt, served through an interactive web app.

AI-Powered Developer Portfolio

- Built a responsive portfolio with React, TypeScript, and Tailwind, featuring an embedded RAG assistant (OpenRouter LLM grounded in my CV with a hallucination guard); deployed on Vercel.

CERTIFICATIONS

- Fundamentals of Deep Learning — NVIDIA (Nov 22, 2022)
- Building Transformer-Based Natural Language Processing Applications — NVIDIA (Dec 19, 2025)
- Applications of AI for Predictive Maintenance — NVIDIA (Oct 27, 2025)

SKILLS

Programming Languages: Python · Java · C · SQL · JavaScript · TypeScript · React.js · React Native · HTML/CSS · Tailwind CSS

AI Libraries & Frameworks: Transformers · PyTorch · TensorFlow · Keras · Hugging Face · Hugging Face Hub · scikit-learn · Pandas · NumPy · Matplotlib

Development Frameworks & Tools: Gradio · Streamlit · Flask · Django · Zeep · LangSmith · Pydantic · Docker Compose · Nginx / Reverse Proxy · Postman · Git · GitHub

Concepts & Technologies: Artificial Intelligence (AI) · Machine Learning (ML) · Deep Learning (DL) · Large Language Models (LLMs) · AI Agents · Agentic AI · Tool-Calling · Prompt Engineering · NLP · ASR · GANs · Vector Databases · Semantic Search · Embeddings · Information Retrieval · API Development · NoSQL · Cloud Computing

EDUCATION

Computer Engineering — AI & Data Science Specialization

2023 – 2026

IEPI – International Polytechnic School · Sousse, Tunisia

Preparatory Cycle

2021 – 2023

EPI – Private International Polytechnic School · Sousse, Tunisia

LANGUAGES

French: DELF B2 **English:** TOEIC (Listening & Reading) — 825/990